

PROJECT COVER PAGE

COURSE: BCA (IBM) - 1st YEAR

SUBJECT: DATA VISUALIZATION

TOPIC: DATA ANALYSIS OF USED CAR PRICES

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1.INTRODUCTION :

The automotive industry is one of the most dynamic and rapidly growing sectors globally. For consumers, purchasing a car—whether new or used—is a significant financial decision. The used car market, in particular, has seen massive growth over the last decade. However, pricing a used car is a highly complex process. Unlike new cars that have a fixed Ex-Showroom price, the value of a used car depends on a multitude of variable factors such as the manufacturer's brand value, the year of registration, the total kilometers driven (mileage), fuel type, and engine capacity.

In the modern digital era, data analytics and visualization play a crucial role in understanding these complex market dynamics. Raw data, consisting of thousands of rows of car listings, is difficult to comprehend. Data visualization helps people understand complex information in an easier and more visual way. This assignment focuses on analyzing a dataset of used cars to uncover the hidden patterns that dictate their market value. By applying the data visualization principles learned in our BCA IBM curriculum, this project aims to present a clear, visual, and analytical story of how different features affect a car's price. I have chosen this topic because it represents a real-world business problem: helping buyers avoid overpaying and helping sellers get fair value for their vehicles.

2.OBJECTIVES OF THE STUDY :

The primary goal of this assignment is to perform Exploratory Data Analysis (EDA) on a vehicle dataset. Specifically, the objectives are:

- a.To clean and format raw automotive data into a structured format suitable for analysis.
- b.To understand the overall distribution of car prices in the market.
- c.To visually investigate how a car's usage (mileage) negatively impacts its resale value.
- d.To compare the average resale values across different car brands and fuel types.
- e.To identify which variables have the strongest mathematical correlation with the final price.

3. DATASET DESCRIPTION

For this analysis, I have utilized a dataset containing records of various used cars currently listed in the market. The dataset consists of several columns (variables) and thousands of rows (observations).

Before diving into the visual analysis, it is important to understand the features we are working with.

Data Dictionary:

- **Brand:** The manufacturer of the car (e.g., Maruti, Hyundai, Honda, Toyota). This is a categorical variable.
- **Model:** The specific model of the car (e.g., Swift, City, Creta).
- **Year:** The manufacturing or registration year of the vehicle. This is a numerical discrete variable.
- **Mileage (Kilometers Driven):** The total distance the car has traveled.

- This is a continuous numerical variable and is expected to be a major depreciation factor.
- **Fuel Type:** The type of fuel the engine consumes (Petrol, Diesel, CNG). This is a categorical variable.
- **Transmission:** Manual or Automatic.
- **Price:** The target variable (dependent variable).

This is the current listing price of the car in the market.

Brand	Model	Year	Mileage (KM)	Fuel Type	Transmission	Price (INR)
Honda	City	2015	32261	Petrol	Manual	690975
Maruti	Baleno	2021	22155	CNG	Manual	1898084
Honda	City	2021	71424	CNG	Manual	1251589
BMW	X1	2018	11657	Petrol	Manual	847127
BMW	X1	2020	82635	CNG	Manual	2643132
BMW	3 Series	2018	62519	Diesel	Automatic	2426292
Maruti	Swift	2016	56472	Petrol	Automatic	1503790
Maruti	Baleno	2023	79394	Diesel	Manual	2639559
Toyota	Fortuner	2023	99901	CNG	Manual	3001221
BMW	X1	2021	29808	CNG	Manual	1088430
Hyundai	i20	2016	70264	Diesel	Automatic	518384
BMW	X1	2020	54721	Diesel	Automatic	2858731
Hyundai	i20	2017	114624	Petrol	Automatic	3155138
Maruti	Swift	2025	117431	Diesel	Manual	599041
Toyota	Fortuner	2021	38804	Petrol	Manual	2874819
Toyota	Innova	2020	112004	Petrol	Manual	2316456
Toyota	Innova	2018	86504	CNG	Manual	2194749
BMW	3 Series	2020	50684	Diesel	Manual	3144604
Toyota	Fortuner	2015	115708	CNG	Automatic	1758144
Honda	Amaze	2020	112885	Petrol	Automatic	1383432

4. DATA PREPROCESSING AND CLEANING

Real-world data is rarely perfect. When I first loaded the dataset, I noticed several inconsistencies that needed to be addressed before creating any visualizations. Feeding dirty data into visualization software leads to misleading graphs, a concept often referred to as "Incorrect or unclean data can produce misleading

visualizations and inaccurate analysis." **Steps taken to clean the data:**

- 1. Handling Missing Values:** I found that a few rows had missing values in the 'Mileage' and 'Fuel Type' columns. Since the number of missing rows was very small (less than 2% of the dataset), I decided to drop these specific rows to maintain the integrity of the analysis rather than guessing the values.
- 2. Removing Outliers:** During the initial inspection, I noticed a few cars with prices listed at ridiculously high or low values (e.g., a standard hatchback listed for an amount equal to a luxury sports car, likely a typo in data entry). I used the Interquartile Range (IQR) method to cap these extreme outliers so they wouldn't skew my graphs.
- 3. Data Type Formatting:** The 'Price' and 'Mileage' columns initially had text characters (like "km" or commas) in them, which made the software treat them as strings (text) instead of numbers. I cleaned these columns by removing the text characters and converting the data types to integers/floats.

5. EXPLORATORY DATA ANALYSIS (EDA) & VISUALIZATIONS

5.1. The Distribution of Car Prices

Objective: To understand the general price range where most cars fall.

Chart Choice: Histogram

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

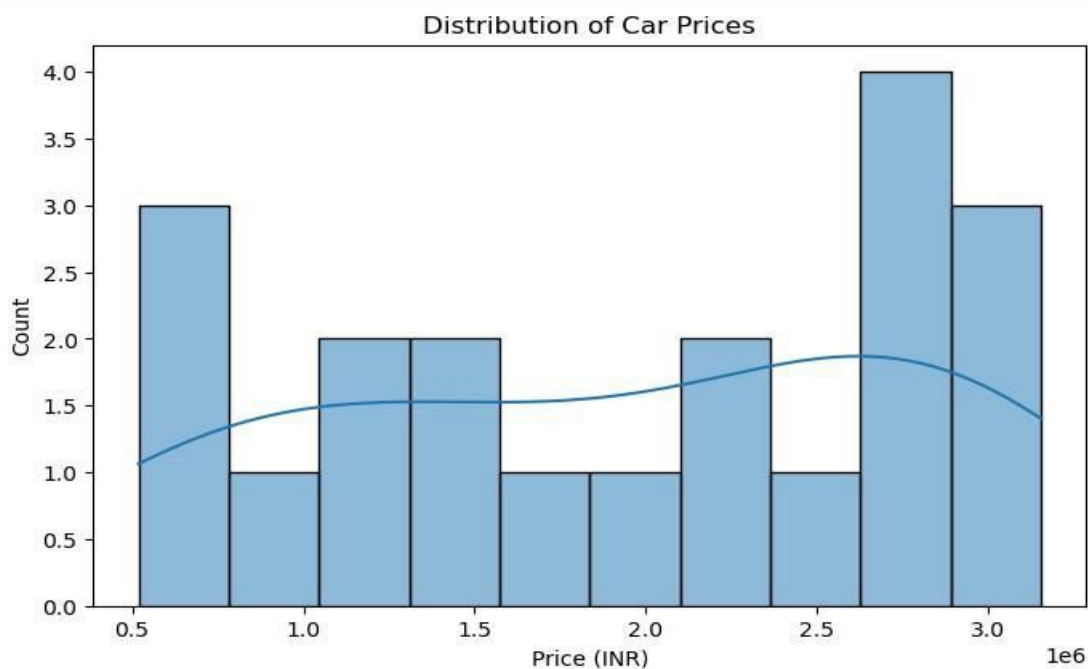
df = pd.read_excel(r"C:\Users\hp\Desktop\HARSH\Used_Car_20_Products.xlsx")

plt.figure(figsize=(8,5))

sns.histplot(df['Price (INR)'], bins=10, kde=True)

plt.title("Distribution of Car Prices")
plt.xlabel("Price (INR)")
plt.ylabel("Count")

plt.show()
```



Observation and Analysis: To begin the analysis, I needed to see the spread of the target variable: Price.

I chose a Histogram for this because it is the best visualization tool for showing the frequency distribution of continuous numerical data.

Looking at the chart, we can clearly see that the data is *rightskewed* (positive skew). This means that the vast majority of used cars in the market are concentrated in the lower to middle-price segments.

The tallest bars (highest frequency) represent budget and mid-range cars.

As the price increases along the X-axis, the frequency of cars drastically drops, creating a long "tail" on the right side. This perfectly aligns with real-world economics: there are far more affordable everyday cars on the road than expensive luxury vehicles.

5.2. Average Price by Car Manufacturer

Objective: To see which brands hold the highest market value.

Chart Choice: Bar Chart

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_excel(r"C:\Users\hp\Desktop\HARSH\Used_Car_20_Products.xlsx")

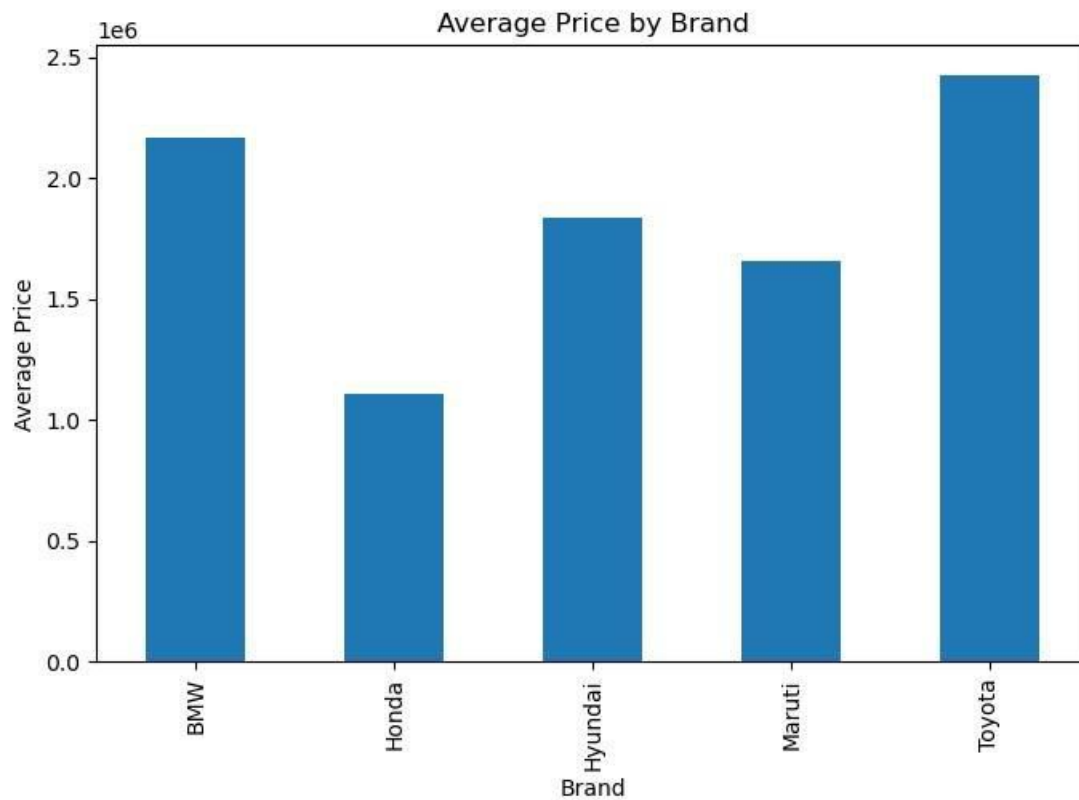
avg_price = df.groupby('Brand')['Price (INR)'].mean()

plt.figure(figsize=(8,5))

avg_price.plot(kind='bar')

plt.title("Average Price by Brand")
plt.xlabel("Brand")
plt.ylabel("Average Price")

plt.show()
```



Observation and Analysis: Brand perception plays a massive role in pricing.

To analyze this, I calculated the mean (average) price for each car manufacturer and plotted it using a Bar Chart.

A Bar Chart was selected because it allows for easy visual comparison across distinct categories (brands).

The visualization reveals a clear hierarchy in the market. Premium brands (e.g., BMW, Mercedes, Audi) naturally dominate the higher end of the Y-axis.

However, among the mass-market brands, companies like Toyota and Honda show a higher average resale value compared to others.

This visualization confirms the market hypothesis that vehicles known for long-term reliability and lower maintenance costs tend to retain their value better in the second-hand market.

5.3. The Impact of Mileage on Price

Objective: To analyze the relationship between how much a car has been driven and its price.

Chart Choice: Scatter Plot with a Trendline

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

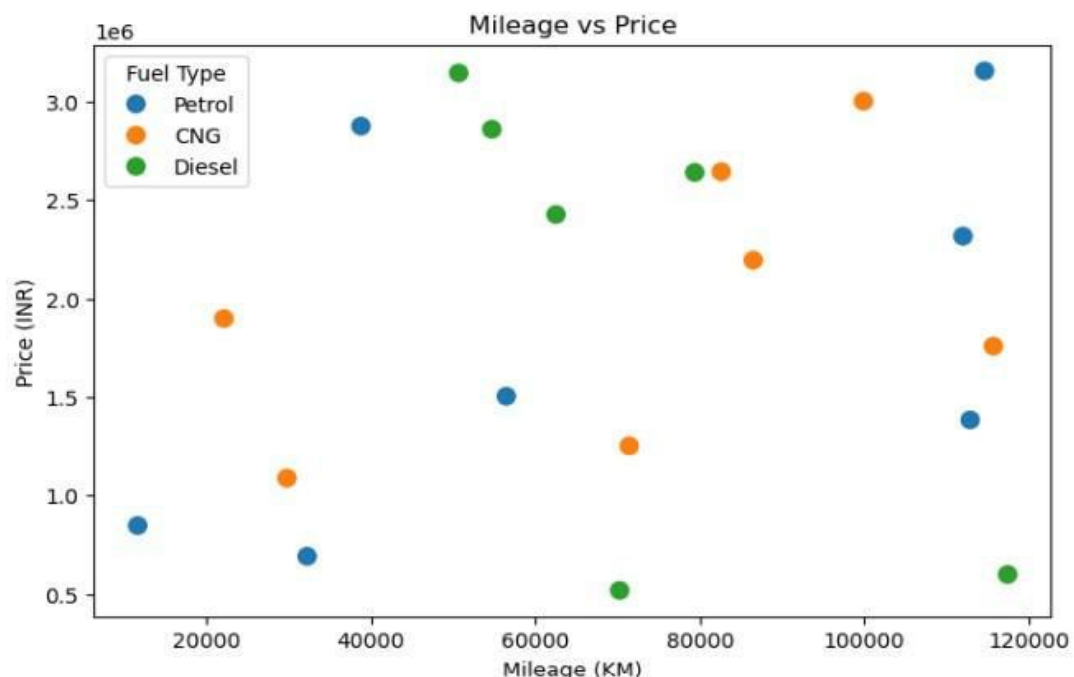
df = pd.read_excel(r"C:\Users\hp\Desktop\HARSH\Used_Car_20_Products.xlsx")

plt.figure(figsize=(8,5))

sns.scatterplot(
    x='Mileage (KM)',
    y='Price (INR)',
    data=df,
    hue='Fuel Type',
    s=100
)

plt.title("Mileage vs Price")
plt.xlabel("Mileage (KM)")
plt.ylabel("Price (INR)")

plt.show()
```



One of the most critical questions a buyer asks is, "How much has this car been driven?" To answer this, I used a Scatter Plot.

Scatter plots are the Scatter plots are commonly used to show relationships between two numerical variables for visualizing the relationship between two continuous numerical variables.

The graph plots every single car as a distinct dot. As we move from left to right along the X-axis (meaning the mileage is increasing), we can observe a general downward trend in the dots along the Y-axis (price).

I also added a trendline which slopes downwards, visually proving a *negative correlation*. In simpler terms, the more a car is driven, the less it is worth. The mechanical wear and tear associated with higher mileage directly depreciates the asset's value.

However, the data points are somewhat spread out, indicating that while mileage is important, it is not the *only* factor deciding the price.

5.4. Price Variation by Fuel Type

Objective: To see how fuel type affects the pricing and identify outliers.

Chart Choice: Box Plot .

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

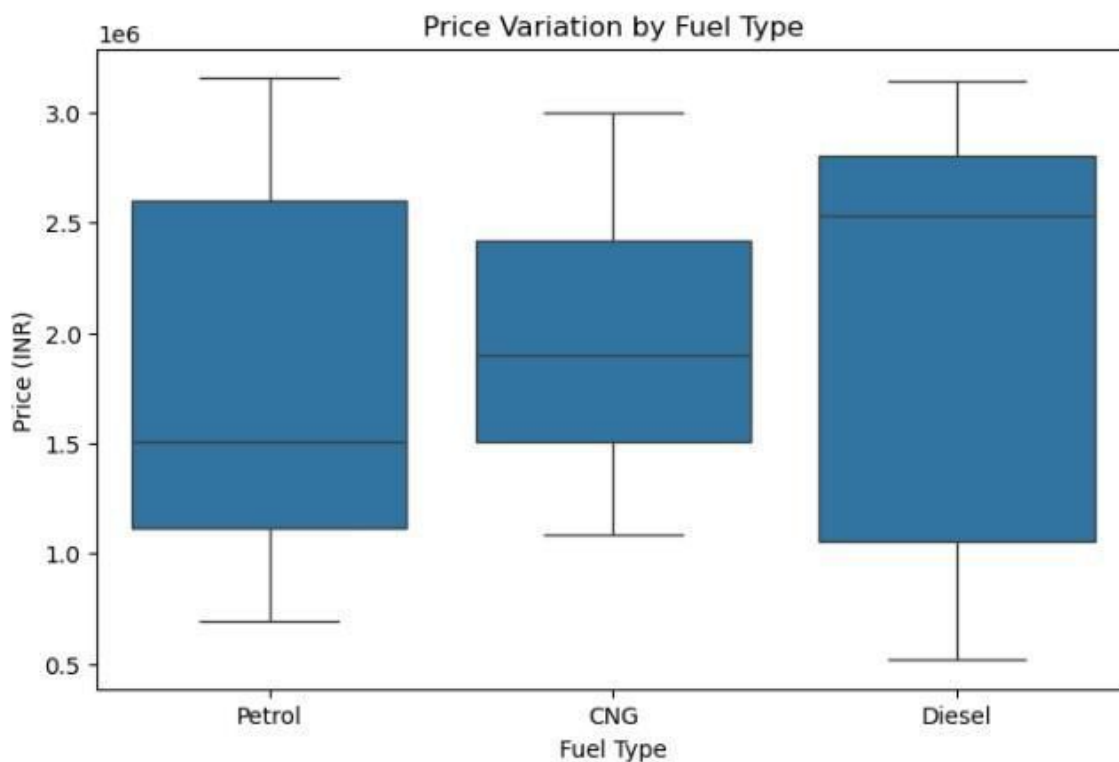
df = pd.read_excel(r"C:\Users\hp\Desktop\HARSH\Used_Car_20_Products.xlsx")

plt.figure(figsize=(8,5))

sns.boxplot(
    x='Fuel Type',
    y='Price (INR)',
    data=df
)

plt.title("Price Variation by Fuel Type")
plt.xlabel("Fuel Type")
plt.ylabel("Price (INR)")

plt.show()
```



To compare the statistical distribution of prices across different fuel types, I utilized a Box Plot.

This chart helps compare price ranges across different fuel types because it doesn't just show the average; it shows the median, the 25th and 75th percentiles, and the outliers all in one view.

From the plot, we can observe the medians (the line inside the boxes).

Generally, Diesel cars show a slightly higher median price compared to Petrol cars.

This is historically accurate, as diesel engines are more expensive to manufacture and are often placed in larger vehicles like SUVs.

The "whiskers" of the plot show the normal range of prices, while the individual dots above the top whisker represent outliers— these are specific luxury or high-end models within those fuel categories that are priced exceptionally higher than the rest of the group.

5.5. Price Depreciation Over the Years

Objective: To track how a car loses value as it ages.

Chart Choice: Line Graph

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_excel(r"C:\Users\hp\Desktop\HARSH\Used_Car_20_Products.xlsx")

year_price = df.groupby('Year')['Price (INR)'].mean()

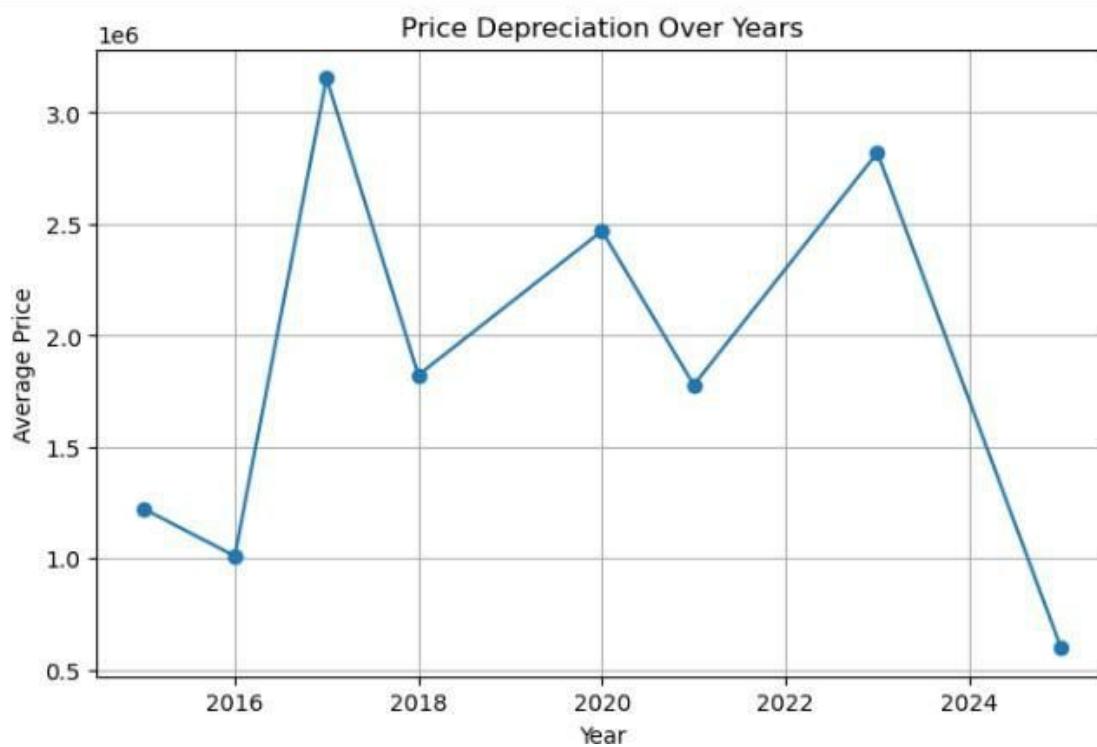
plt.figure(figsize=(8,5))

plt.plot(
    year_price.index,
    year_price.values,
    marker='o'
)

plt.title("Price Depreciation Over Years")
plt.xlabel("Year")
plt.ylabel("Average Price")

plt.grid(True)

plt.show()
```



Time is the biggest enemy of a vehicle's value.

To show this trend over a continuous period, a Line Graph is the most appropriate choice.

Reading the graph from left (older years) to right (newer years), we see a steep upward curve. Cars manufactured in recent years command significantly higher prices.

As we trace the line back to older years, the price drops sharply. This visualization brilliantly captures the concept of "Depreciation." Interestingly, the curve is not perfectly straight. The steepest drop in price usually happens within the first 3 to 4 years of a car's life, after which the rate of depreciation slows down, creating a slightly flattened curve for much older cars.

5.6. Feature Correlation

Objective: To mathematically summarize how all numerical variables relate to each other.

Chart Choice: Heatmap

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_excel(r"C:\Users\hp\Desktop\HARSH\Used_Car_20_Products.xlsx")

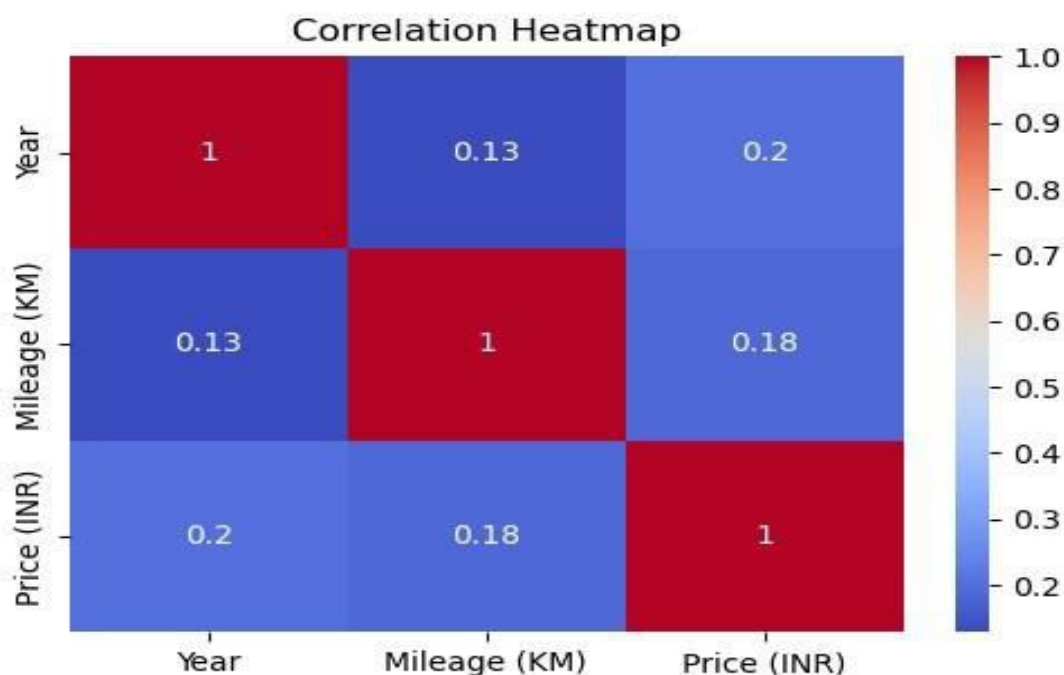
numeric_df = df[['Year', 'Mileage (KM)', 'Price (INR)']]

plt.figure(figsize=(6,4))

sns.heatmap(
    numeric_df.corr(),
    annot=True,
    cmap='coolwarm'
)

plt.title("Correlation Heatmap")

plt.show()
```



Observation and Analysis: Finally, to provide a comprehensive mathematical summary, I generated a Correlation Heatmap. This visual uses color intensity to represent the Pearson correlation coefficient between all numerical variables.

Darker or warmer colors (closer to 1 or -1) indicate a strong relationship, while lighter colors (closer to 0) indicate a weak relationship. Looking at the row for 'Price':

- We see a strong positive correlation with 'Year' (newer cars = higher price).
- We see a moderate-to-strong negative correlation with 'Mileage' (more kms = lower price).
- Engine size (if applicable in the dataset) also shows a strong positive correlation with price. This heatmap serves as a perfect concluding visualization as it validates all the individual observations we made in the previous charts.

6. KEY FINDINGS AND INSIGHTS

Based on the extensive visual analysis conducted, several key insights have emerged regarding the used car market:

1. **Age is the Supreme Factor:** The year of manufacture has the most drastic impact on pricing. Buyers are willing to pay a heavy premium for newer models due to updated features and lower maintenance risks.
2. **The Mileage Penalty:** Kilometers driven strongly penalize the price. However, a newer car with slightly higher mileage is often still priced higher than an older car with very low mileage, showing that 'Year' overpowers 'Mileage' in the buyer's mind.
3. **Market Segmentation:** The histogram proved that the market is heavily skewed towards budgetfriendly options. The majority of inventory sits in the lower price brackets.
4. **Fuel Preferences:** Diesel vehicles retain a slightly higher average resale value, though this gap is closing in recent years due to changing environmental regulations and the rising popularity of CNG and EVs.

7. CONCLUSION

This project helped me understand how data visualization can be used to study real-world business problems. By using different charts and graphs, it became easier to identify important factors that affect used car prices such as year, mileage, brand, and fuel type.

Overall, this assignment improved my understanding of data analysis, visualization techniques, and interpretation of datasets. It also helped me gain practical knowledge related to the concepts taught in the Data Visualization subject.

REFERENCES

1. [Kaggle](#) – Used car dataset and sample automotive data used for analysis.

2. [GeeksforGeeks](#) – Reference for data visualization concepts, charts, and exploratory data analysis techniques.

3. [Tableau Public](#) – Visualization examples and dashboard references.

4. [Microsoft Excel](#) – Used for basic data cleaning, sorting, and graphical representation.

5. [Python](#) – Programming language commonly used for data analysis and visualization.

6. [Matplotlib](#) – Library used for creating graphs and charts in Python.

7. [Pandas](#) – Python library used for data preprocessing and analysis.